A 12-wk egg intervention increases serum zeaxanthin and macular pigment optical density in women.

Two carotenoids found in egg yolk, lutein and zeaxanthin, accumulate in the macular retina where they may reduce photostress. Increases in serum lutein and zeaxanthin were observed in previous egg interventions, but no study measured macular carotenoids. The objective of this project was to determine whether increased consumption of eggs would increase retinal lutein and zeaxanthin, or macular pigment. Twenty-four females, between 24 and 59 y, were assigned to a pill treatment (PILL) or 1 of 2 egg treatments for 12 wk. Individuals in the PILL treatment consumed 1 sugar-filled capsule/d. Individuals in the egg treatments consumed 6 eggs/wk, containing either 331 microg (EGG 1) or 964 microg (EGG 2) of lutein and zeaxanthin/yolk. Serum cholesterol, serum carotenoids, and macular pigment OD (MPOD) were measured at baseline and after 4, 8, and 12 wk of intervention. Serum cholesterol concentrations did not change in either egg treatment group, but total cholesterol ($P = 0.04$) and triglycerides ($P = 0.02$) increased in the PILL group. Serum zeaxanthin, but not serum lutein, increased in both the EGG 1 ($P = 0.04$) and EGG 2 ($P = 0.01$) groups. Likewise, MPOD increased in both the EGG 1 ($P = 0.001$) and EGG 2 ($P = 0.049$) groups. Although the aggregate concentration of carotenoid in 1 egg yolk may be modest relative to other sources, such as spinach, their bioavailability to the retina appears to be high. Increasing egg consumption to 6 eggs/wk may be an effective method to increase MPOD.